Advanced Computational Physics, University of Strathclyde, WS 2026,
Dr. Ben Hourahine

Assignment 4

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1 ISING MODEL

1.1 Metropolis

The results of applying the standard Metropolis scheme to the Ising model are shown in Figure 3 for lattices of size $L = 16, 64, 256, 1024$. We expect a phase transition of second order to happen at the critical temperature $T_c = 2/\ln(1+\sqrt{2}) \approx 2.269$. Thus, the system should be in the ordered state for $T < T_c$, i.e. $M/L^2 = 1$, and the linear response C_V should diverge at $T = T_c$. Due to considering a finite system, the specific heat does not diverge but spike. For $L = 16$ such could be observed. For larger lattices, the algorithm did not fully converge in $16 \cdot 1000 \cdot L^2$. Reason for this is that each Metropolis step only flips one spin, making it almost impossible to reach the fully ordered state in reasonable time. This issue will be adressed later on. An immediate fix would be to intialize the lattice in the ordered state, however this will break down for $T > T_c$ and the XY model.

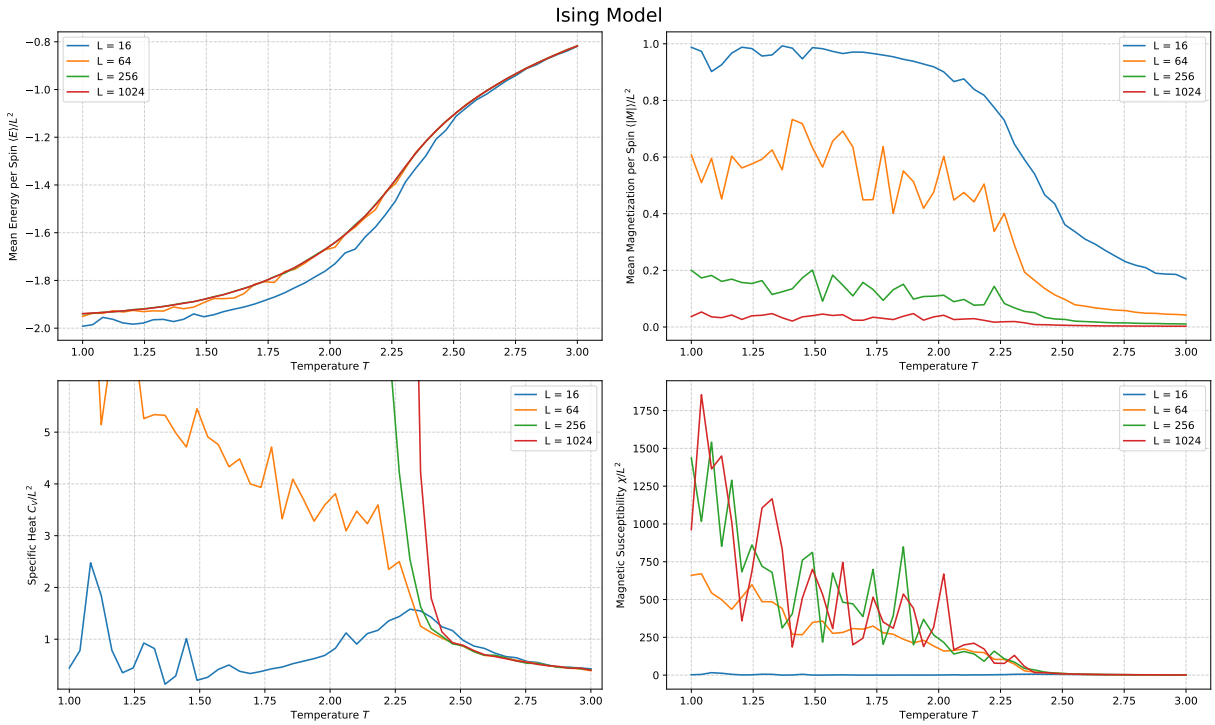

Figure 1

Further we explore the parallelism of the algorithm using *Amdahls* law. This is shown in figure 2, yielding a close to one parallelism $p = 0.9954$ and inflection point at $n_{inf} = 271.1 \approx 2^{7.76}$.

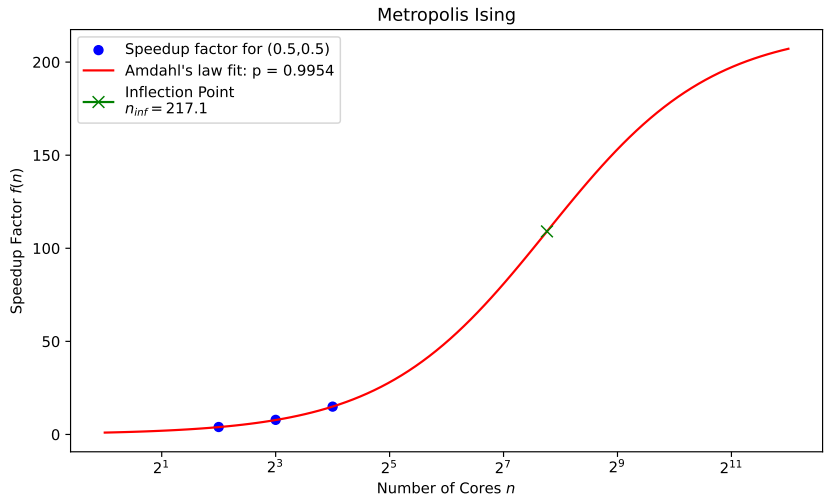

Figure 2

1.2 Wolff

To handle larger lattices the Wolff algorithm was implemented, flipping entire clusters. This also corresponds to a Metropolis walk, however now steps are used to determine the lattice and its size to flip. The results of applying this scheme to the Ising model are shown in Figure **??** for lattices of size $L = 16, 64, 256, 1024$. Observe, that now the ordered phase appears as expected, aswell as a peak in the specific heat.

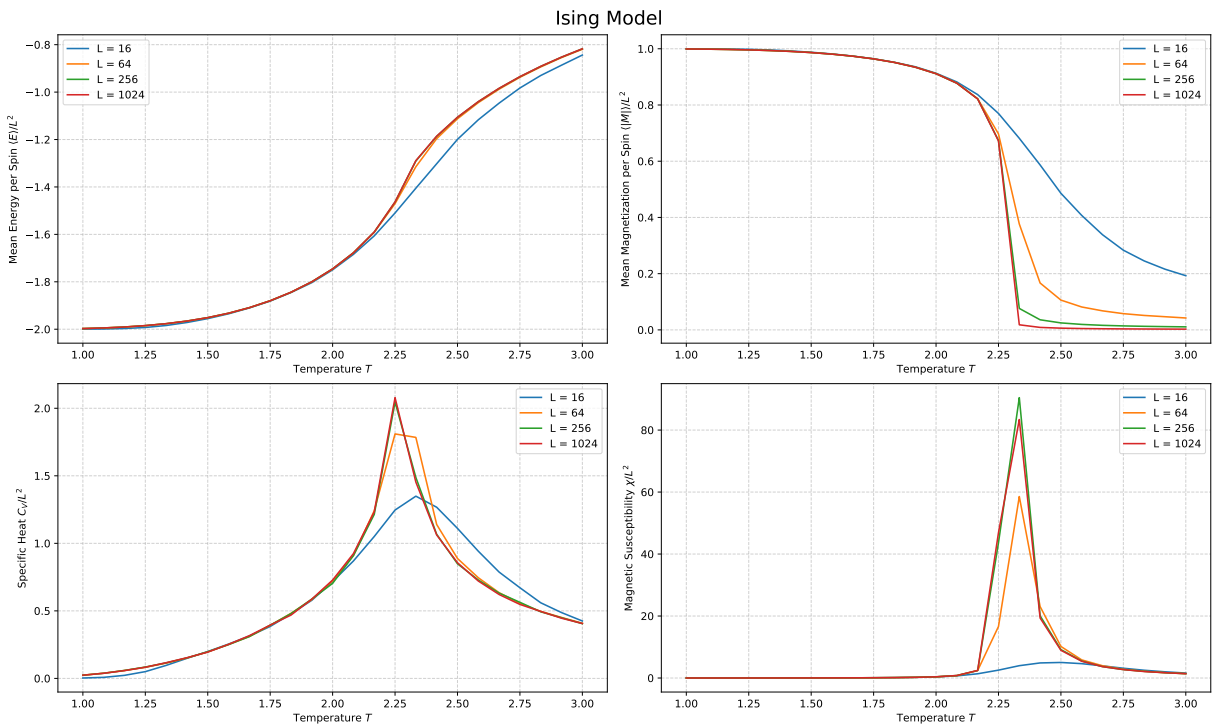

Figure 3

Again *Amdahls* law yields a close to one parallelism $p = 0.9957$ and inflection point at $n_{inf} = 233.3 \approx 2^{7.78}$.

2 XY MODEL

2.1 Metropolis

The XY model in two dimensions exhibits a BKT phase transition at $T_c = 0.893$. This means that below T_c a ordered phse appears forming vortices. We will see, that a quasi long range order appears, given by a power law behavior of correlations. In the high temperature phase, i.e. disorder, a exponential decay appears. We still expect a maximum of the specific heat at T_c. Results obtained with standard Metropolis are shown in Figure 4. Observe, that for $L = 16$ the mentioned expectations appear. Be-

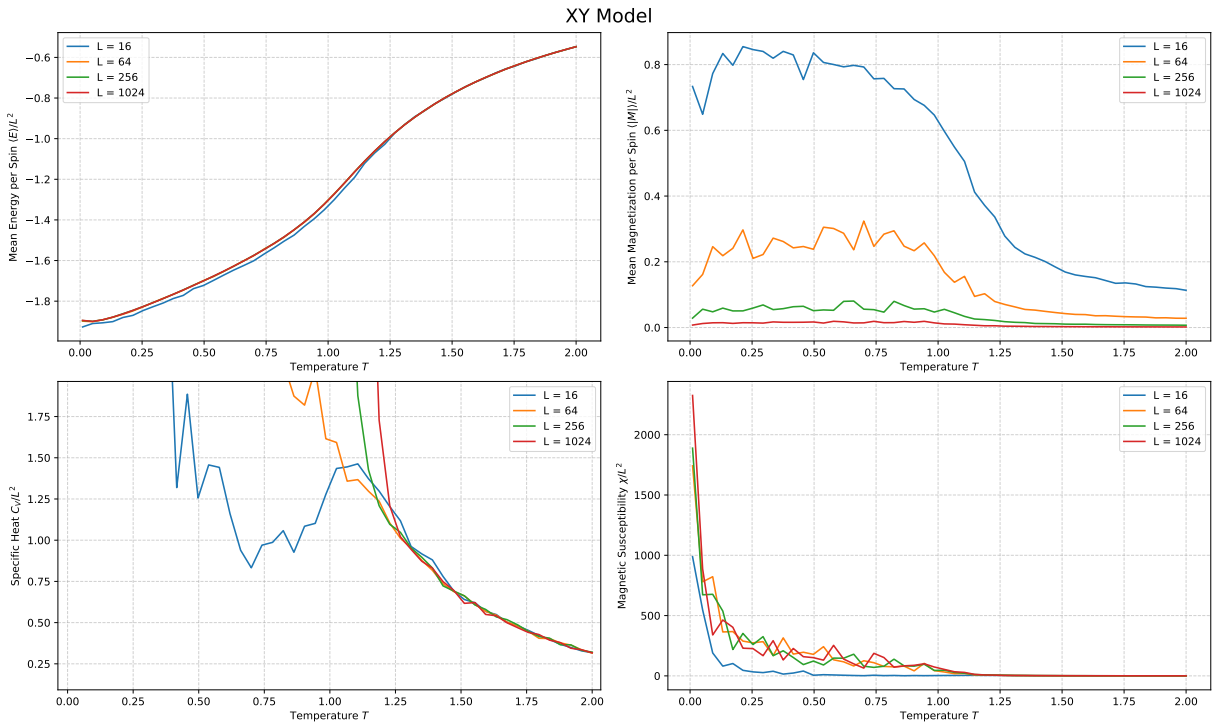

Figure 4

yond that the system cannot find its equilibrium state in reasonable time. Note, that the specific heat diverges due to being energy fluctuations divided by temperature, only saturating if the equilibrium state is found.

Again *Amdahls* law yields a close to one parallelism $p = 0.9974$ and inflection point at $n_{inf} = 382.0 \approx 2^{8.58}$.

2.2 Wolff

Again a Wolff cluster algorithm was implemented to handle larger lattices. The results are shown in Figure 5. These match the predictions better.

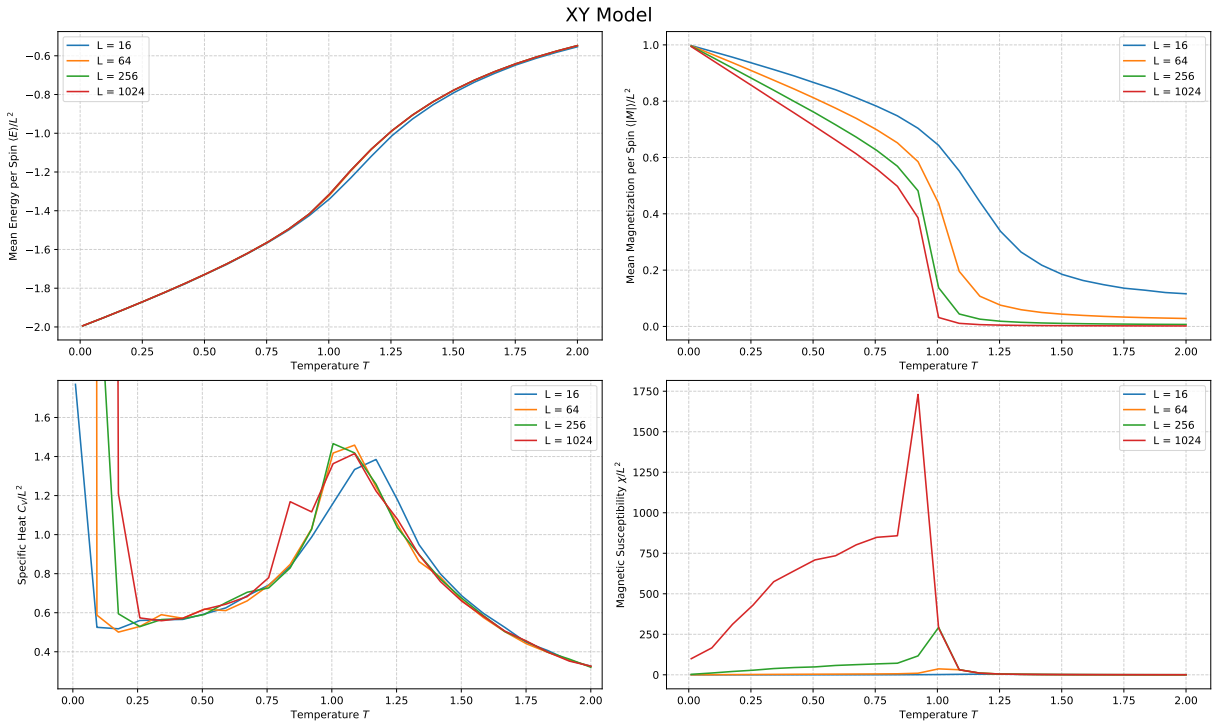

Figure 5

Again *Amdahls* law yields a close to one parallelism $p = 0.9985$ and inflection point at $n_{inf} = 682.0 \approx 2^{9.4}$.

2.3 Spin Correlations

We expect a power law behavior below the critical temperature and an exponential decay above. The correlation functions for $L = 1024$ and $T = 0.5, 2.0$ are shown in figure 6. The better converging Wolff algorithm was used to obtain these correlations.

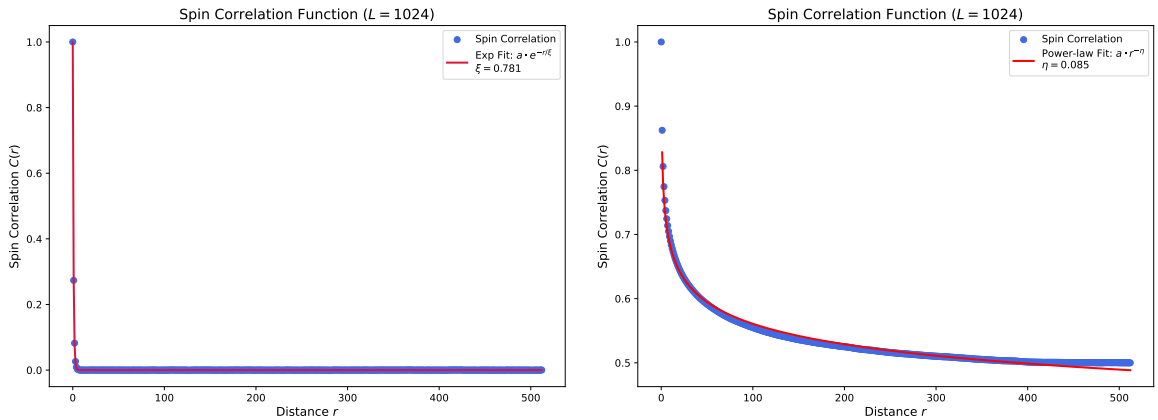

Figure 6

Find the repository under `https://github.com/joern16/advanced-computational-physics`.